Gene Essentiality from the Cell Graph

The backbone against flux balance analysis and Flux Cone Learning,

and what routing the prediction through the metabolic module does

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Abstract

Gene essentiality is the one endpoint on which a flux model has a published, non-trivial score: on the released Merzbacher 2025 yeast split, flux balance analysis reads an AUROC of 0.69 over all 1,121 labeled genes and Flux Cone Learning 0.73 on the 223 held-out genes. This document scores the cell graph transformer on the same held-out genes, puts FBA on every medium the ontology defines, and asks two questions of the metabolic module: whether a prediction read off its growth column ranks essential genes better than the backbone alone, and where the gradient goes when the module is one of two paths rather than the only one. Every number here is at a validation-selected epoch on a pinned test set, with a gene bootstrap and the across-seed range, and every table is generated from the result files.

Section status: ✓ ready ■ author-reviewed, keep stable ✗ not done, agents may edit freely

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1 The endpoint, its labels and its splits ✗

1.1 Vocabulary ✗

AUROC Area under the receiver operating characteristic, computed as the Mann-Whitney rank statistic with mid-ranks on ties: the probability that a randomly drawn essential gene scores above a randomly drawn nonessential one.

backbone The cell graph transformer over the nine-graph perturbed gene graph with `prot_T5_all` node features, read at the pooled perturbed tokens. The *pooled* arm is this backbone with a scalar head and no metabolic module.

gene bootstrap Resampling the scored genes with replacement (2,000 draws) and reporting the 2.5 and 97.5 percentiles of the statistic.

held-out genes The 223 genes Merzbacher 2025 held out of its own training (`test = 1` in the released split), of which 195 resolve as single-deletion genotypes in the build. Every model number in this document is on those 195 unless the gene set is named.

metabolic module The enzyme-constrained thermodynamic flux layer over yeast-GEM 9.0.2 that maps the perturbed representation to a flux vector; here it is read at the objective column `r_2111` (growth).

released labels The 1,121 essentiality calls in the released split. The column named `essential` is a viability flag (1.0 marks a nonessential gene); it is inverted once and the four class counts are asserted against the paper's Methods.

selected epoch The argmax of a centered five-epoch moving average of the validation AUROC. The held-out genes are scored once, at that epoch.

1.2 Where the labels come from ✗

The build is a union of the SGD essential-gene list (1,140 genotypes, all essential) and the single-mutant fitness screens (Costanzo 2016, Kuzmin 2018; 4,540 genotypes), 5,656 single-deletion genotypes in all. No fitness value enters any objective: the head is one logit scored by binary cross entropy against a 0/1 label, and every genotype whose gene carries no label arrives masked. Two label sources are used and named throughout:

Table 1: The split files, one per label source: genes per split with the essential count in parentheses. The test block is the released held-out genes under either source; validation is a class-stratified fifth of the remaining labeled genes at seed 0. Under the genome source the released label wins on every disagreement (12 genes).

split file	labels	train (ess)	val (ess)	test (ess)
`ess_fcl_val0.2_s0`	released (1,121)	629 (100)	158 (25)	195 (31)
`ess_genome_val0.2_s0`	released + SGD essential + viable fitness (5,795)	4,368 (879)	1,093 (220)	195 (31)

1.3 The protocol inherited from 027, and what it got wrong ✗

The wave-9 protocol pinned the held-out genes and let the data module draw train and validation at 80/10/10 over the remaining 5,461 records. Of those, 926 carry a label, so the validation set held 75 labeled genes with about 12 essential, on which a single AUROC read has a Hanley-McNeil standard error near 0.07; 10 percent of the labeled genes sat in a random test block nothing reports; and 4,535 unlabeled records were forwarded every epoch under a False mask. A split file replaces the draw: it names every record's split, is class-stratified, carries the labels it was stratified on, and the trainer drops every record it does not name.

2 The comparators, computed rather than quoted ✗

2.1 Flux balance analysis on four media ✗

The published FBA comparator is cobrapy `single_gene_deletion` with glucose as the carbon source, scored on all 1,121 labeled genes. The phenotype it is compared against was called on rich medium, and the model here is scored on the 195 held-out genes, so the comparison as published is neither on the same medium nor on the same genes. Table 2 recomputes it on yeast-GEM 9.0.2 under the exchange bounds the model ships with and under the three media the ontology defines, with the essentiality score being the negative knockout-to-wild-type growth ratio, rounded to 10^{-3} so

that the roughly 900 deletions that grow at 0.99 to 1.00 of the wild type rank as the tie they are (without the rounding two identical runs differed by 0.03 AUROC on the held-out genes).

Table 2: FBA single-gene deletion on yeast-GEM 9.0.2 (cobrapy 0.30.0), scored as the negative knockout-to-wild-type growth ratio rounded to 10^{-3}. AUROC on all 1,121 labeled genes, on the 223 held-out genes, and on the 195 held-out genes the model scores with a 2,000-draw gene bootstrap interval. 1121 of the labeled genes are in the GEM; the rest keep wild-type growth.

medium	WT growth (1/h)	lethal	all 1,121	held-out 223	our 195 [95% CI]
shipped bounds (glucose minimal, aerobic)	0.086	186	0.707	0.633	0.654 [0.56, 0.75]
SM, synthetic minimal	0.314	160	0.707	0.642	0.666 [0.57, 0.76]
SC, synthetic complete	0.543	91	0.719	0.653	0.666 [0.57, 0.76]
YPD approximation (no peptone)	0.535	104	0.715	0.653	0.664 [0.57, 0.76]

The medium moves the number of lethal deletions by a factor of two and the AUROC by about 0.01. The published 0.69 on all labeled genes is reproduced to within 0.02.

2.2 Flux Cone Learning on the same genes ✗

Recomputed from the authors' released per-gene predictions: 0.7308 on the 223 held-out genes (published 0.73) and 0.742 on the 195 genes the model scores. On those same 195 genes the released FBA predictions score 0.640, so the published pair overstates FBA by about 0.06 relative to a head-to-head on one gene set. The paper prints a third number, 0.78 ± 0.065 in the legend of its Figure 3a, which is the mean over five cross-validation folds drawn inside its 898 training genes; a model scored on the held-out genes is compared with the 0.73 line, never with that one.

2.3 The classical ladder on the same split ✗

The fitness and expression lines each carry a ladder of models that need no graph and no training beyond a solve, and essentiality gets the classification form of it: FBA's growth ratio with no fitting, then four model classes (L_2 logistic regression, a random forest, a radial-basis support vector classifier, cosine nearest neighbors) on each of twenty-four gene encodings, every pair fit twice, with and without the four FBA growth ratios appended. The encodings are the seventeen the fitness ladder carries (codon frequency, the chromosome and pathway indicator, six nucleotide windows, the two FUDT embeddings, the species language model, CaLM, ProtT5 and ESM2 with and without dubious genes, and their concatenation, `mixed`) plus seven controls: a one-hot gene identity, and standard normal vectors of dimension 1, 10, 100, 1,000, 6,607 (the one-hot's width) and 8,192, drawn once at seed 0. A one-hot rung must score chance on held-out genes, since every held-out column is zero in training, and a random vector must score chance at every width; either scoring above chance would mean the split leaks. Every fitted rung is fit on the split file's 629 training genes, its one hyperparameter chosen by five-fold stratified cross-validation inside those genes, and scored once on the 158 validation and 195 held-out genes, on which nothing was chosen. The transformer cells in Sections 3 and 4 train on the same 629 genes and score on the same 195. Figure 1 shows all 192 fitted rungs; Table 3 keeps, per encoding, the model class whose cross-validation AUROC is highest.

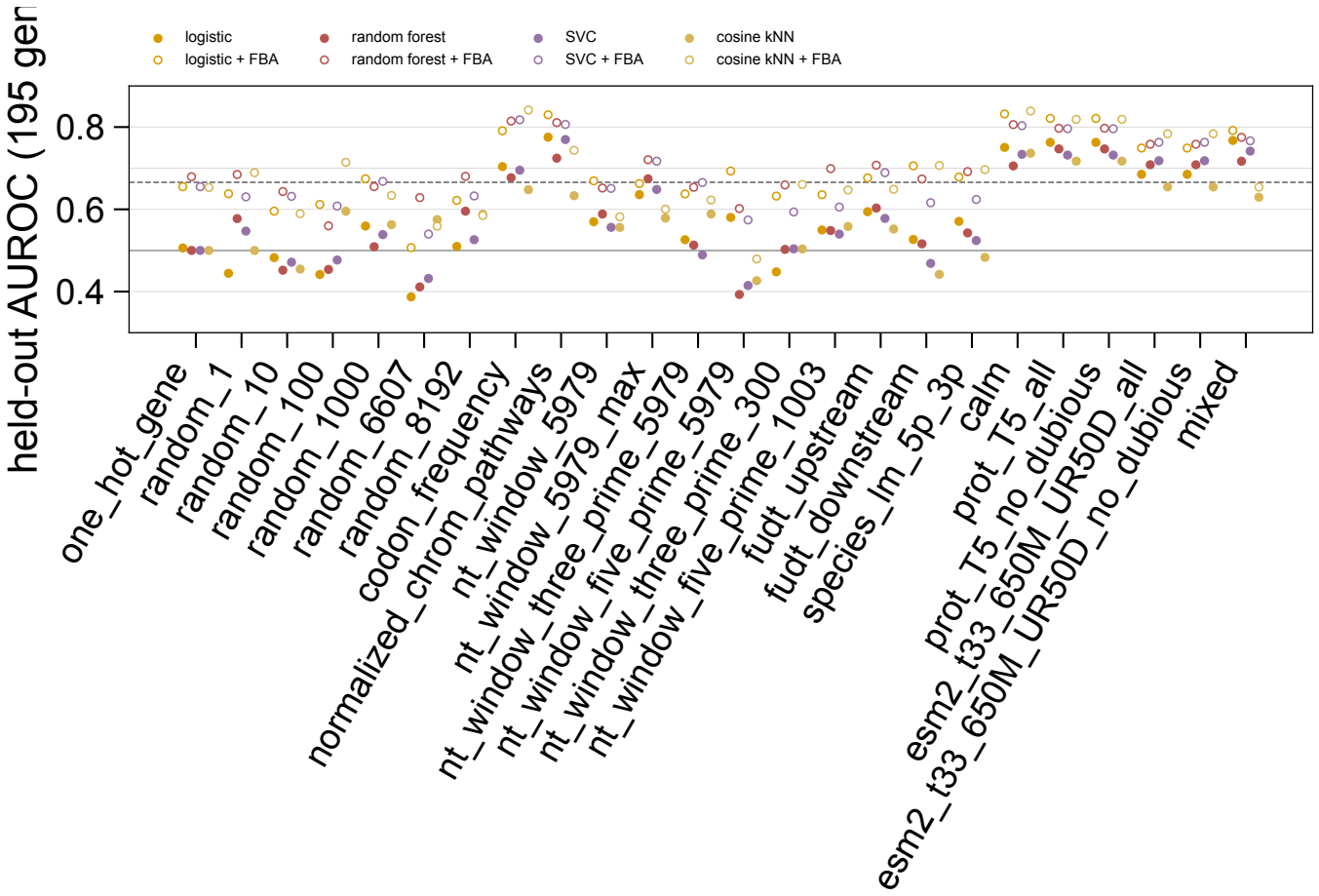


Figure 1: The classical ladder on the released-label split. Held-out AUROC on the 195 genes for every model class (color) on every encoding (column), filled without and open with the four FBA growth ratios appended; the dashed line is FBA alone on the shipped medium (0.654), the solid line chance. The hyperparameter of each mark was chosen by five-fold cross-validation inside the 629 training genes. Written by `experiments/028-gene-essentiality/scripts/logistic_baselines.py`.

Table 3: The classical ladder on the released-label split (`ess_fcl_val0.2_s0.json.gz`). Per encoding, the model class whose five-fold cross-validation AUROC inside the 629 training genes is highest, with and without FBA's four growth ratios appended; the validation genes and the 195 held-out genes (31 essential) are scored once, on genes no fit touched, the held-out number with a 2,000-draw gene bootstrap. FBA alone is not fitted. Every model and encoding is in Figure 1.

encoding	best model (chosen)	dim	CV	val	held-out [95% CI]
FBA growth	medium=shipped	–	–	0.683	0.654 [0.56, 0.75]
FBA growth	medium=SM	–	–	0.710	0.666 [0.57, 0.76]
FBA growth	medium=SC	–	–	0.726	0.666 [0.57, 0.76]
FBA growth	medium=YPD	–	–	0.714	0.664 [0.57, 0.76]
one_hot_gene	logistic (C=10)	6,579	0.504	0.508	0.506 [0.50, 0.52]
one_hot_gene + FBA	random forest (max_depth=8)	6,583	0.775	0.730	0.679 [0.59, 0.77]
random_1	random forest (max_depth=4)	1	0.598	0.560	0.577 [0.46, 0.69]
random_1 + FBA	cosine kNN (k=10)	5	0.806	0.699	0.689 [0.56, 0.80]
random_10	cosine kNN (k=50)	10	0.549	0.481	0.455 [0.35, 0.56]
random_10 + FBA	logistic (C=0.001)	14	0.786	0.684	0.596 [0.46, 0.72]
random_100	random forest (max_depth=None)	100	0.522	0.376	0.454 [0.34, 0.57]
random_100 + FBA	random forest (max_depth=4)	104	0.773	0.662	0.560 [0.42, 0.70]
random_1000	random forest (max_depth=4)	1,000	0.475	0.490	0.509 [0.40, 0.62]
random_1000 + FBA	random forest (max_depth=None)	1,004	0.778	0.749	0.655 [0.53, 0.78]
random_6607	cosine kNN (k=1)	6,607	0.549	0.477	0.575 [0.49, 0.66]
random_6607 + FBA	random forest (max_depth=None)	6,611	0.734	0.713	0.628 [0.50, 0.75]
random_8192	cosine kNN (k=50)	8,192	0.523	0.515	0.588 [0.49, 0.69]
random_8192 + FBA	random forest (max_depth=8)	8,196	0.768	0.748	0.680 [0.57, 0.79]
codon_frequency	SVC (C=1)	64	0.700	0.663	0.695 [0.59, 0.79]
codon_frequency + FBA	SVC (C=1)	68	0.849	0.772	0.818 [0.73, 0.90]
normalized_chrom_pathways	logistic (C=10)	197	0.705	0.722	0.775 [0.68, 0.86]
normalized_chrom_pathways + FBA	logistic (C=0.01)	201	0.802	0.791	0.830 [0.75, 0.90]
nt_window_5979	logistic (C=0.01)	2,560	0.579	0.532	0.570 [0.46, 0.67]
nt_window_5979 + FBA	logistic (C=0.01)	2,564	0.743	0.705	0.669 [0.55, 0.79]
nt_window_5979_max	logistic (C=0.001)	2,560	0.626	0.608	0.636 [0.54, 0.73]
nt_window_5979_max + FBA	logistic (C=0.01)	2,564	0.777	0.786	0.663 [0.54, 0.78]
nt_window_three_prime_5979	logistic (C=0.001)	2,560	0.555	0.580	0.526 [0.41, 0.64]
nt_window_three_prime_5979 + FBA	logistic (C=0.001)	2,564	0.732	0.720	0.637 [0.51, 0.76]
nt_window_five_prime_5979	cosine kNN (k=25)	2,560	0.594	0.518	0.427 [0.32, 0.54]
nt_window_five_prime_5979 + FBA	logistic (C=0.01)	2,564	0.733	0.657	0.693 [0.59, 0.79]
nt_window_three_prime_300	random forest (max_depth=4)	2,560	0.564	0.537	0.503 [0.39, 0.62]
nt_window_three_prime_300 + FBA	random forest (max_depth=None)	2,564	0.752	0.757	0.659 [0.53, 0.78]
nt_window_five_prime_1003	logistic (C=0.001)	2,560	0.541	0.485	0.550 [0.43, 0.68]
nt_window_five_prime_1003 + FBA	logistic (C=0.01)	2,564	0.727	0.665	0.636 [0.52, 0.75]
fudt_upstream	SVC (C=1)	768	0.611	0.583	0.578 [0.47, 0.69]
fudt_upstream + FBA	random forest (max_depth=None)	772	0.775	0.747	0.707 [0.60, 0.81]
fudt_downstream	SVC (C=10)	768	0.589	0.553	0.469 [0.36, 0.58]
fudt_downstream + FBA	logistic (C=0.1)	772	0.770	0.735	0.706 [0.59, 0.81]
species_lm_5p_3p	logistic (C=0.001)	1,536	0.634	0.631	0.571 [0.47, 0.68]
species_lm_5p_3p + FBA	logistic (C=0.01)	1,540	0.763	0.733	0.679 [0.56, 0.79]
calm	SVC (C=1)	768	0.755	0.777	0.734 [0.64, 0.82]
calm + FBA	random forest (max_depth=8)	772	0.850	0.833	0.806 [0.72, 0.89]
prot_T5_all	SVC (C=10)	1,024	0.781	0.767	0.732 [0.62, 0.83]
prot_T5_all + FBA	logistic (C=0.01)	1,028	0.842	0.826	0.821 [0.74, 0.90]
prot_T5_no_dubious	SVC (C=10)	1,024	0.781	0.767	0.732 [0.62, 0.83]
prot_T5_no_dubious + FBA	logistic (C=0.01)	1,028	0.842	0.826	0.821 [0.74, 0.90]
esm2_t33_650M_UR50D_all	random forest (max_depth=8)	1,280	0.731	0.694	0.708 [0.61, 0.80]
esm2_t33_650M_UR50D_all + FBA	random forest (max_depth=8)	1,284	0.815	0.794	0.758 [0.66, 0.86]
esm2_t33_650M_UR50D_no_dubious	random forest (max_depth=8)	1,280	0.731	0.694	0.708 [0.61, 0.80]
esm2_t33_650M_UR50D_no_dubious + FBA	random forest (max_depth=8)	1,284	0.815	0.794	0.758 [0.66, 0.86]
mixed	logistic (C=10)	6,149	0.788	0.794	0.768 [0.66, 0.87]
mixed + FBA	logistic (C=10)	6,153	0.833	0.837	0.792 [0.69, 0.89]

The controls behave. The one-hot rung scores 0.506 and the six random widths score 0.45 to 0.59, chance within the 195-gene noise, and with FBA appended they score 0.56 to 0.69, FBA's own number, so no rung reads anything but its features. Without FBA the protein and codon language models are the only encodings clear of 0.70: ProtT5 0.732, CaLM 0.734, the concatenation 0.768, and the chromosome-pathway indicator 0.775 [0.68, 0.86], each level with Flux Cone Learning's 0.742 on the same genes. FBA lifts every one of them: logistic regression on ProtT5 with the ratios scores 0.821 [0.74, 0.90] from 1,029 parameters, the support vector classifier on 64 codon frequencies 0.818 [0.73, 0.90], the random forest on CaLM 0.806 [0.72, 0.89], and logistic regression on the 197-dimensional chromosome-pathway indicator 0.830 [0.75, 0.90]. The rung with the highest cross-validation AUROC, the random forest on CaLM with FBA at 0.850, scores 0.806 on the held-out genes; the highest held-out number in the figure, 0.841 for cosine nearest neighbors on codon frequency with FBA, is the maximum over 192 rungs and is not a selected model. Neither an embedding nor the solver reaches 0.80 alone, and their sum does on four encodings. FBA's ratio is a feature the ontology computes

for every gene at no training cost, so the transformer is measured against 0.80 to 0.83 from here on, not against the published 0.73.

3 The backbone: splits, labels, weight decay and the shape of the curve ✗

Every cell in this section is the pooled backbone at width 12 with one transformer block and a one-layer preprocessor (16,874 parameters), trained for 80 epochs at three initialization seeds with the split pinned. Round 1 (jobs 2149 to 2181) moves one lever at a time away from the cell that trains on the released-label split file (`ess_fcl_val0.2_s0`, 629 training genes, 158 validation genes): the old 75-gene validation draw, the genome-wide labels, weight decay at 0, 10^{-2} and 10^{-1}, batch 32, learning rate at 3×10^{-4} and 3×10^{-3}, the class weight 772/126, and graph regularization off. Table 4 carries every number; Figure 2 the curves.

Table 4: Round 1, the backbone. Width 12, one block, one-layer preprocessor, 80 epochs, seeds 2700 to 2702. Held-out AUROC on the 195 genes: the seed mean with its standard deviation, the three seeds, the gene bootstrap of the pooled per-gene logits, the epochs selected on validation, and the median absolute epoch-to-epoch step of the validation curve.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
wave-11 protocol (75-gene val)	0.759	0.020	0.781, 0.755, 0.741	0.760 [0.67, 0.84]	18, 38, 14	0.0056
released labels, 158-gene val	0.725	0.019	0.745, 0.722, 0.708	0.738 [0.65, 0.83]	66, 77, 77	0.0026
weight decay 0	0.725	0.019	0.745, 0.720, 0.709	0.735 [0.64, 0.82]	66, 77, 77	0.0027
weight decay 1e-2	0.726	0.017	0.745, 0.722, 0.711	0.735 [0.64, 0.83]	65, 77, 77	0.0029
weight decay 1e-1	0.725	0.018	0.746, 0.721, 0.709	0.735 [0.64, 0.82]	66, 77, 77	0.0025
batch 32	0.765	0.005	0.766, 0.770, 0.759	0.764 [0.67, 0.85]	74, 73, 77	0.0032
genome labels (5,795 genes)	0.704	0.024	0.706, 0.726, 0.679	0.705 [0.61, 0.80]	77, 77, 69	0.0020
fcl_lr3e-3	0.686	0.043	0.718, 0.704, 0.637	0.713 [0.62, 0.81]	77, 77, 50	0.0024
fcl_lr3e-4	0.658	0.093	0.737, 0.682, 0.556	0.732 [0.64, 0.82]	77, 77, 74	0.0023
fcl_noreg	0.725	0.019	0.745, 0.722, 0.709	0.738 [0.65, 0.83]	66, 77, 77	0.0026
fcl_posw	0.759	0.016	0.764, 0.772, 0.741	0.769 [0.68, 0.86]	77, 75, 53	0.0034

3.1 The curve was noisy because the validation set was small ✗

The median epoch-to-epoch step of the validation AUROC is 0.0056 on the 75-gene draw, 0.0026 on the 158-gene split and 0.0020 on the 1,093-gene genome-wide split (Figure 2b). Two of the three width-24 cells of the previous protocol selected epochs 4 and 5 on validation peaks of 0.82 and 0.86 and scored 0.74 on the held-out genes; no cell on a split file selects before epoch 50. The old protocol's curve reads near 0.8 on its own 75 genes and the split cells read 0.65 to 0.72 on theirs, so the levels are not comparable, only the steps are.

3.2 The split file costs training genes, and the cells are under-trained ✗

The split-file cell scores 0.725 against the old protocol's 0.759 on the same held-out genes. It trains on 629 labeled genes where the old draw trained on roughly 740, and every one of its seeds selected epoch 66 to 77 of 80 with the validation curve still rising (Figure 2a). At batch 128 an epoch is five optimizer steps, so 80 epochs is 400 updates. The two levers that move all three seeds in the same direction are both more optimization: batch 32, four times the steps per epoch, lifts the seeds by 0.020, 0.048 and 0.051 and collapses their spread to 0.005; the class weight, which steepens the curve, lifts them by 0.019, 0.051 and 0.033. Round 3 combines the two and raises the epoch budget.

3.3 Weight decay and graph regularization do nothing here ✗

Weight decay at 0, 10^{-2} and 10^{-1} reproduces the base cell to the third decimal on every seed, and so does removing graph regularization. At 400 updates with learning rate 10^{-3}, a decay of 10^{-1} shrinks a weight by four percent; the graph term's λ is 6.5×10^{-5}. Neither is a lever on this endpoint at this budget. The learning rate at 3×10^{-4} and 3×10^{-3} is worse than 10^{-3} on the seed mean, with seed spreads of 0.09 and 0.04 that put the pooled numbers inside the base interval.

3.4 More labels from the genome make the held-out ranking worse ✗

The genome-wide file adds SGD's essential list and the viable single-mutant fitness records as nonessential, 4,368 training genes against 629, and scores 0.704 against 0.725, three of three seeds down. Its validation AUROC on its own 1,093

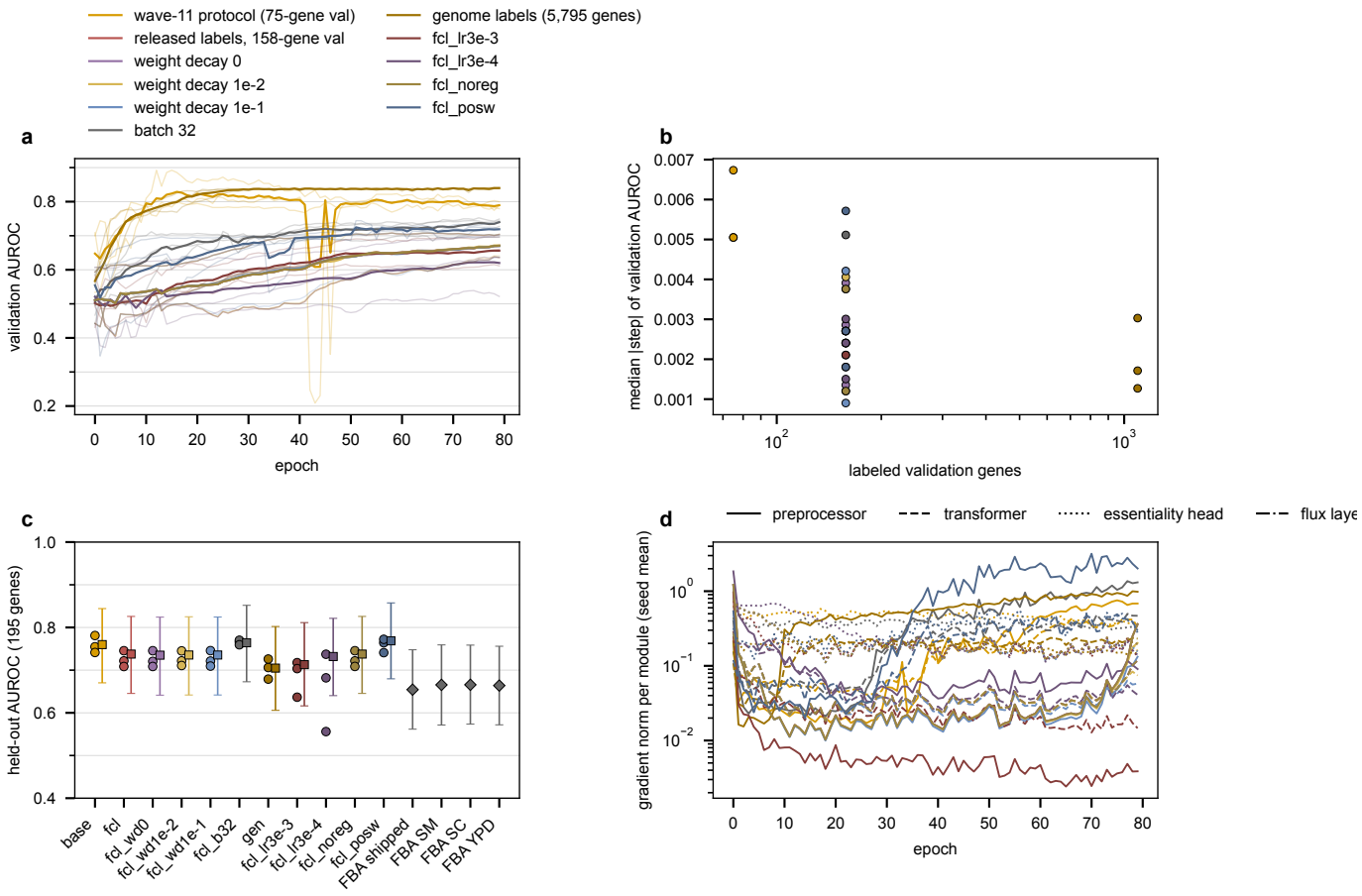


Figure 2: Round 1, the backbone. **a**, Validation AUROC per epoch, thin lines per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step of that curve against the number of labeled validation genes it was read on. **c**, Held-out AUROC on the 195 genes per cell, the seed mean, and the gene bootstrap of the pooled per-gene logits, beside FBA on each medium. **d**, Per-module gradient norm per epoch, seed mean, on the arms that logged it. Generated by `scripts/wave12_readout.py -round w12`.

genes is 0.84 while the released held-out genes read 0.70, the widest validation-to-held-out gap of any arm. The viable-fitness negatives are a different population from the released split's nonessential genes, and a model that ranks them well does not rank the released genes better. Hypothesis, untested: the released split's nonessential class is enriched for genes with a measurable fitness defect, which the genome-wide negatives are not.

3.5 Gradient norms ✗

On the split-file cells the preprocessor's gradient norm starts near 10^{-2} and rises about thirty-fold over 80 epochs while the head's stays flat near 0.4 (Figure 2d); the clip at 10 never binds. The rise tracks the validation curve's climb, which is the same under-training reading from the gradient side.

3.6 Round 3: the two levers combined, a longer budget, and the fold ensemble ✗

Round 3 (jobs 2209 and 2224 to 2236) combines the two round-1 levers that moved three of three seeds, batch 32 and the class weight, at 80 and at 200 epochs, runs batch 32 alone at 200 epochs as the class weight's control, and trains five fold models on the cross-validation files, which partition the 787 labeled non-test genes five ways with the held-out genes fixed; their logits are averaged on the held-out genes, the protocol behind the published Flux Cone Learning number (Table 5, Figure 3). With the loader workers removed a 200-epoch cell takes 11 minutes.

Table 5: Round 3, more optimization and the fold ensemble. Batch 32 with the class weight at 80 and 200 epochs, batch 32 alone at 200 epochs, and the five fold models of the cross-validation files, each on its own fold.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
batch 32 + class weight, 80 epochs	0.750	0.043	0.780, 0.769, 0.700	0.774 [0.69, 0.86]	75, 68, 31	0.0026
batch 32 + class weight, 200 epochs	0.774	0.006	0.780, 0.769, 0.773	0.780 [0.69, 0.87]	75, 68, 148	0.0031
batch 32, 200 epochs	0.761	0.005	0.766, 0.761, 0.755	0.764 [0.67, 0.85]	74, 100, 181	0.0026
5-fold, fold 0	0.780	0.000	0.780	0.780 [0.69, 0.86]	75	0.0033
5-fold, fold 1	0.806	0.000	0.806	0.806 [0.72, 0.89]	50	0.0036
5-fold, fold 2	0.797	0.000	0.797	0.797 [0.72, 0.87]	14	0.0033
5-fold, fold 3	0.751	0.000	0.751	0.751 [0.66, 0.84]	184	0.0021
5-fold, fold 4	0.789	0.000	0.789	0.789 [0.70, 0.87]	108	0.0039

Batch 32 with the class weight at 200 epochs scores 0.780, 0.769 and 0.773, a seed mean of 0.774 with a spread of 0.006, the tightest cell of any round; the pooled bootstrap is 0.780 [0.69, 0.87]. Batch 32 alone at 200 epochs scores 0.766, 0.761 and 0.755, so the class weight adds 0.01 to 0.02 on three of three seeds at the longer budget as it did at 80. The longer budget itself moves one seed: the 80- and 200-epoch cells share a trajectory through epoch 80 and select the same epoch on seeds 2700 and 2701 (75 and 68), while seed 2702 selects epoch 31 at 80 epochs and epoch 148 at 200, 0.700 against 0.773. Fold 0 of the cross-validation files is the released-label split file exactly (same 629 training, 158 validation and 195 held-out genes, both drawn at seed 0), so its fold model is the 200-epoch cell's first seed. The five fold models score 0.780, 0.806, 0.797, 0.751 and 0.789 on the held-out genes, a fold mean of 0.785, and their averaged logits score 0.795 [0.71, 0.88] (`cv_ensemble_readout.py`). That is the number to set beside the published 0.73 and beside 0.742 on these genes: the backbone under Flux Cone Learning's own protocol is 0.05 above it, and 0.01 to 0.04 below the ladder's rungs with FBA appended.

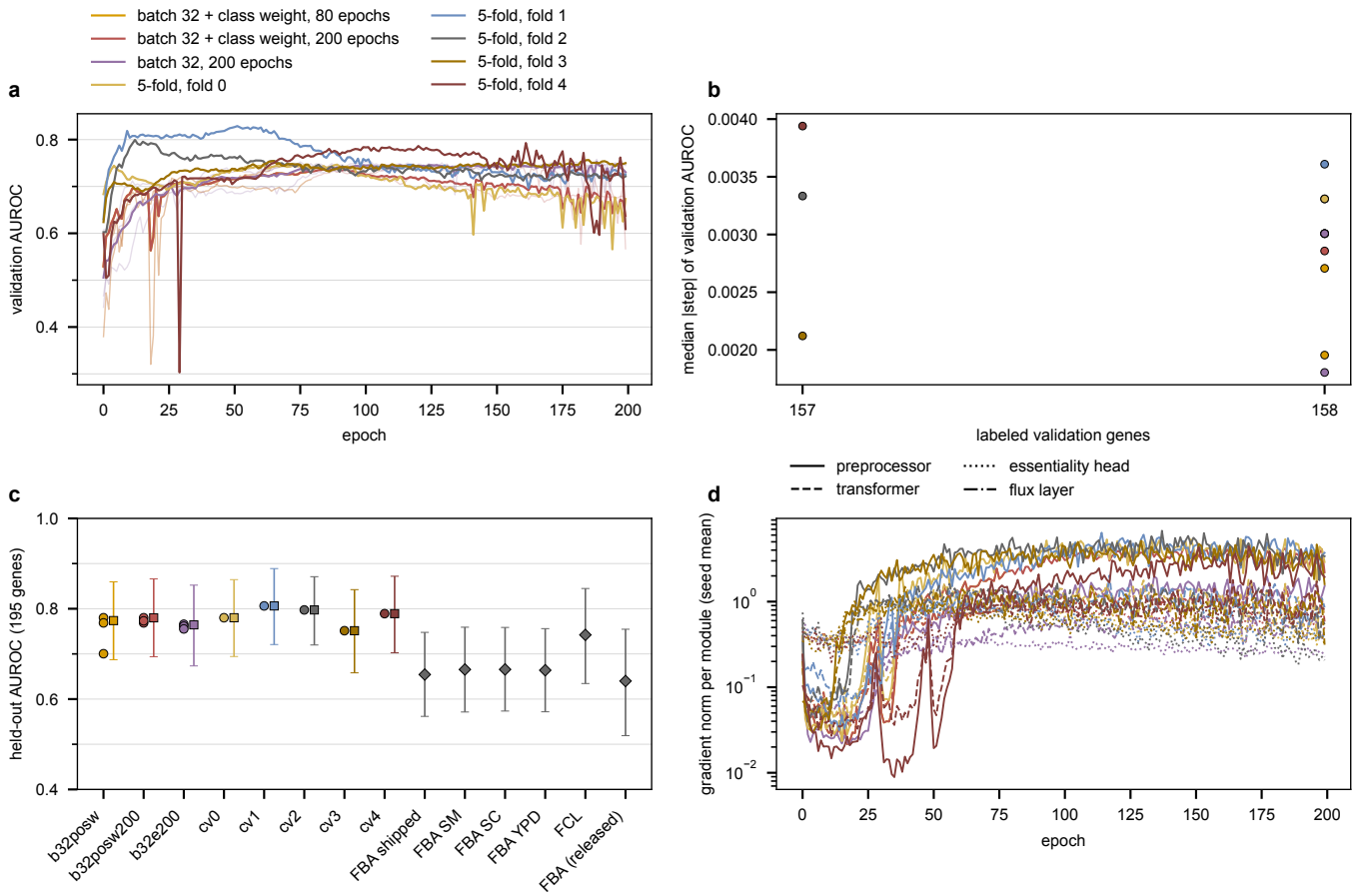


Figure 3: Round 3, the levers combined and the fold models. **a**, Validation AUROC per epoch, seed mean per arm; each fold model reads its own validation fold. **b**, The median absolute epoch-to-epoch step per cell against the validation size, 157 or 158 genes. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w14`.

4 The metabolic module: three ways to read essentiality off it ✗

The pooled backbone reads its logit from an MLP on the perturbed context (Figure 4). A flux arm builds the enzyme-constrained thermodynamic flux layer on the same context and exposes one physically named quantity, the flux through the growth reaction `r_2111`. Three readouts put that quantity to different uses, and they differ in what the backbone's gradient has to pass through.

biomass (forced) $\ell = a\, v_{\text{growth}} + b$. The growth column is the only path from the context to the logit, so the module carries the whole prediction or the cell scores chance (Figure 5). This is FBA's own reading of essentiality with a learned threshold, and it is the wave-9 design.

context + growth $\ell = \text{MLP}(h) + a\, v_{\text{growth}} + b$. Both paths from the first step; the per-module gradient norms say how much of each update the module carries when it is not forced to carry all of it.

gated $\ell = \text{MLP}(h) + g\,(a\, v_{\text{growth}} + b)$ with $g_0 = 0$ (Figure 6). At initialization the cell is the pooled model exactly; the gate's trajectory, logged every epoch, says whether the optimizer chose the flux path, and the `flux_layer` gradient line says whether gradient reached the layer, which at $g = 0$ it cannot.

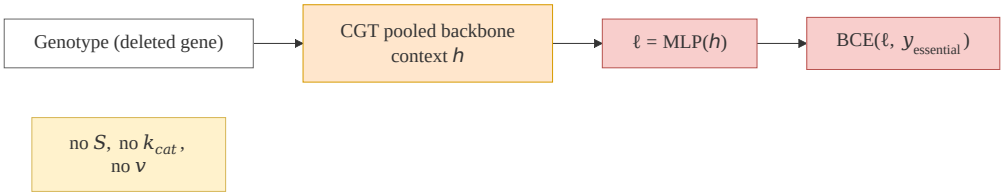


Figure 4: The pooled readout. The backbone alone: one MLP from the perturbed context to a logit, binary cross entropy against the released label. No stoichiometry, no kinetic parameters, no flux. Rendered from `notes/experiments.028-gene-essentiality.mermaid.readouts.md`.

4.1 Round 2: the module on the stable backbone, with and without the penalties ✗

Round 2 (jobs 2182 to 2208) puts the three readouts on the width-12 cell of Section 3, on the released-label split, at three seeds each: the forced and gated readouts at constraint scale 0.1 and 0, the context readout at both scales, the forced readout at scale 0 with the k_{cat} vector permuted across catalytic units (same architecture, no unit-to-value correspondence), and the stoichiometry-only layer (`flux_off`: no thermodynamics, no capacity) with the forced and context readouts. Every flux arm carries the wave-9 flux configuration otherwise: the predicted k_{cat} tier, the union-external thermodynamic source, twenty warmup epochs, the dataset medium, the mixture concentration space. The paired comparator is the round-1 pooled cell on the same split at the same batch, weight decay and budget (Table 6, Figure 7).

The reads were fixed before the numbers: *not hurt* is the gated arm against the pooled cell paired by seed, with the gate's trajectory beside it; *help* is the forced arm above the pooled cell on three of three seeds and above its shuffled-k_{cat} control; the *collapse's cause* is the forced arm at scale 0.1 against scale 0, since on the betaxanthin targets the penalty scale was the dose that starved the layer's input.

Table 6: Round 2, the metabolic module. The round-1 pooled cell on the released-label split (`fc1`) and the same cell with the prediction routed through the flux layer: the growth column alone (forced), the context head plus the growth affine, and that affine behind a zero-initialized gate, at constraint scale 0.1 and 0, with the shuffled-kcat control and the stoichiometry-only layer.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
released labels, 158-gene val	0.725	0.019	0.745, 0.722, 0.708	0.738 [0.65, 0.83]	66, 77, 77	0.0026
growth column only, constraints 0.1	0.706	0.105	0.759, 0.585, 0.774	0.765 [0.67, 0.85]	77, 26, 76	0.0037
growth column only, constraints 0	0.732	0.068	0.770, 0.653, 0.772	0.773 [0.68, 0.86]	72, 72, 72	0.0047
growth column only, kcat shuffled, constraints 0	0.751	0.029	0.771, 0.718, 0.765	0.769 [0.68, 0.86]	74, 77, 75	0.0029
context + growth column, constraints 0.1	0.666	0.086	0.566, 0.709, 0.722	0.717 [0.61, 0.82]	49, 77, 77	0.0053
context + growth column, constraints 0	0.667	0.096	0.556, 0.727, 0.718	0.724 [0.62, 0.82]	49, 73, 77	0.0068
gated growth column, constraints 0.1	0.719	0.029	0.721, 0.747, 0.690	0.738 [0.64, 0.83]	41, 77, 77	0.0027
gated growth column, constraints 0	0.735	0.018	0.714, 0.744, 0.747	0.747 [0.65, 0.84]	40, 77, 77	0.0031
growth column only, stoichiometry only	0.606	0.072	0.524, 0.631, 0.661	0.664 [0.56, 0.77]	19, 73, 65	0.0033
context + growth column, stoichiometry only	0.705	0.023	0.719, 0.717, 0.678	0.755 [0.67, 0.84]	75, 76, 61	0.0042

Not hurt holds. The gated arm at scale 0 scores 0.714, 0.744 and 0.747 against the pooled cell's 0.745, 0.722 and 0.708 on the same seeds, two of three above by 0.02 to 0.04 and one below by 0.03, a seed mean of 0.735 against 0.725; at

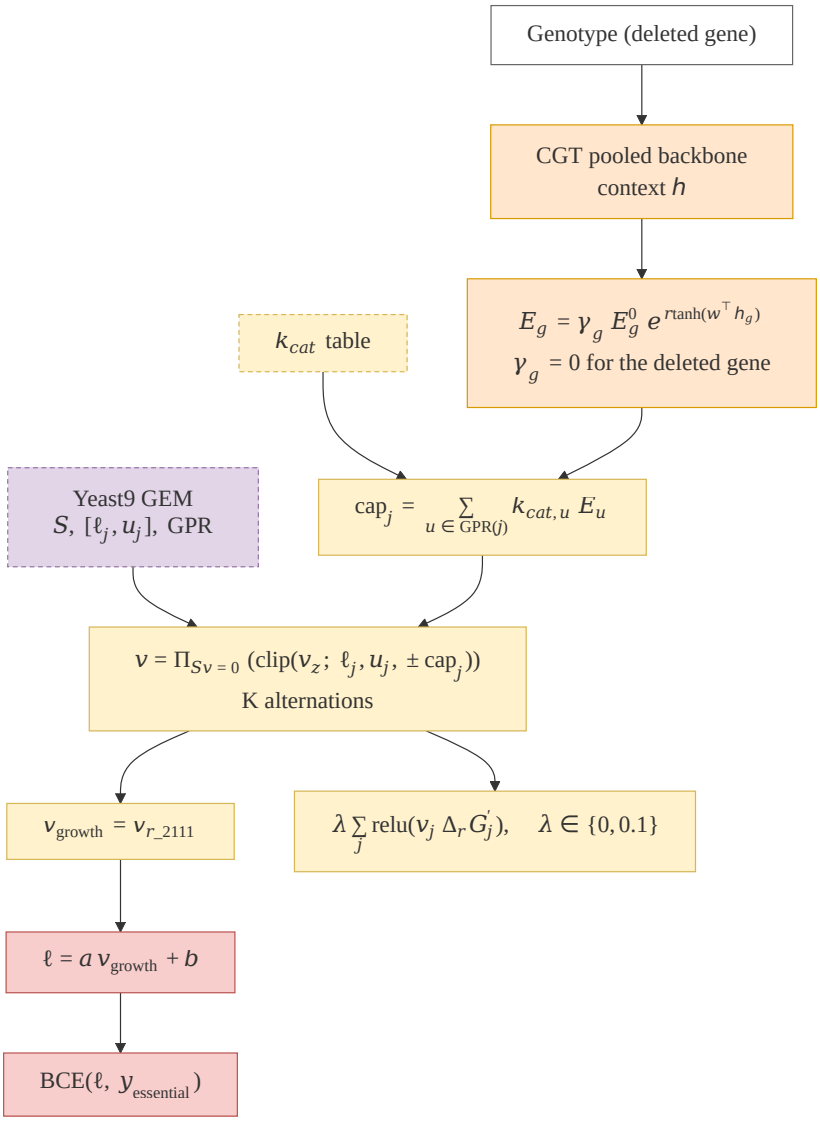


Figure 5: The forced biomass readout. The context sets each enzyme's abundance (a deleted gene's is zero), the k_{cat} table turns abundances into capacities, and the flux is the projection onto $Sv = 0$ of the capacity-clipped proposal; the logit is an affine of the growth column alone. The thermodynamic penalty is a loss on the flux with no path to the prediction, at scale 0.1 or 0.

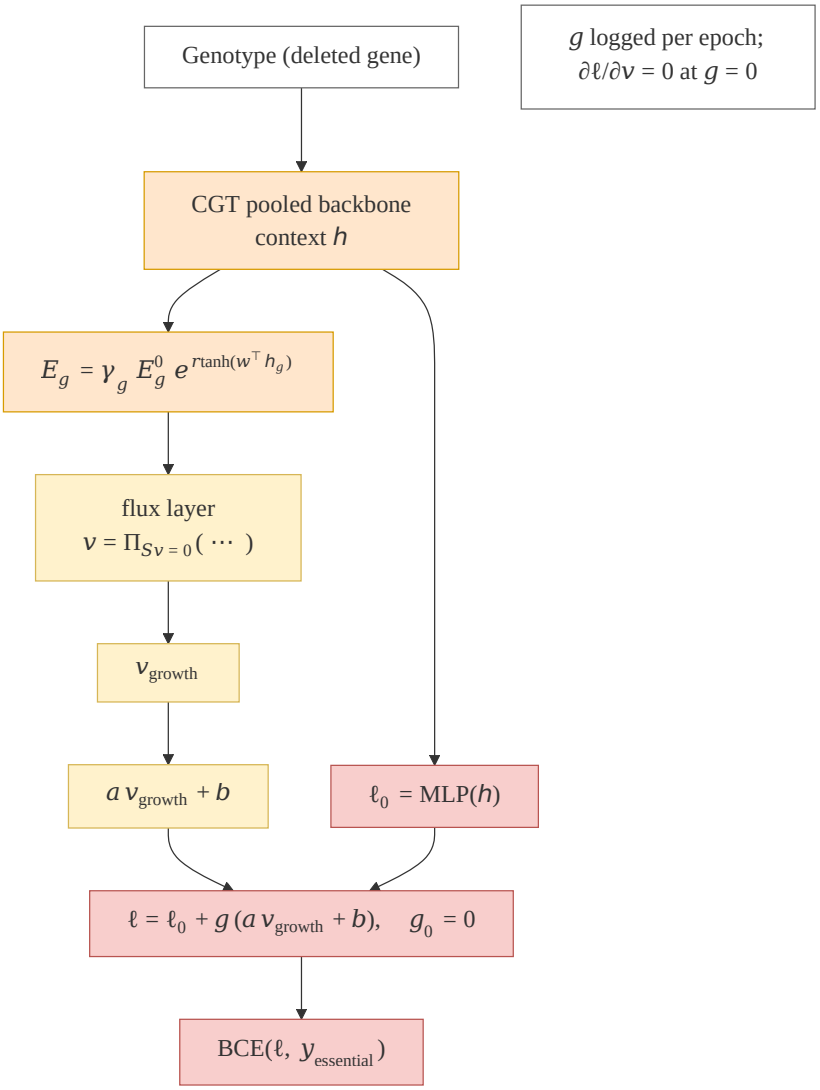


Figure 6: The gated readout. The pooled MLP's logit plus the growth affine behind a gate initialized at zero. The gate is a parameter of the readout; its value is written into every epoch row as `ess_gate`.

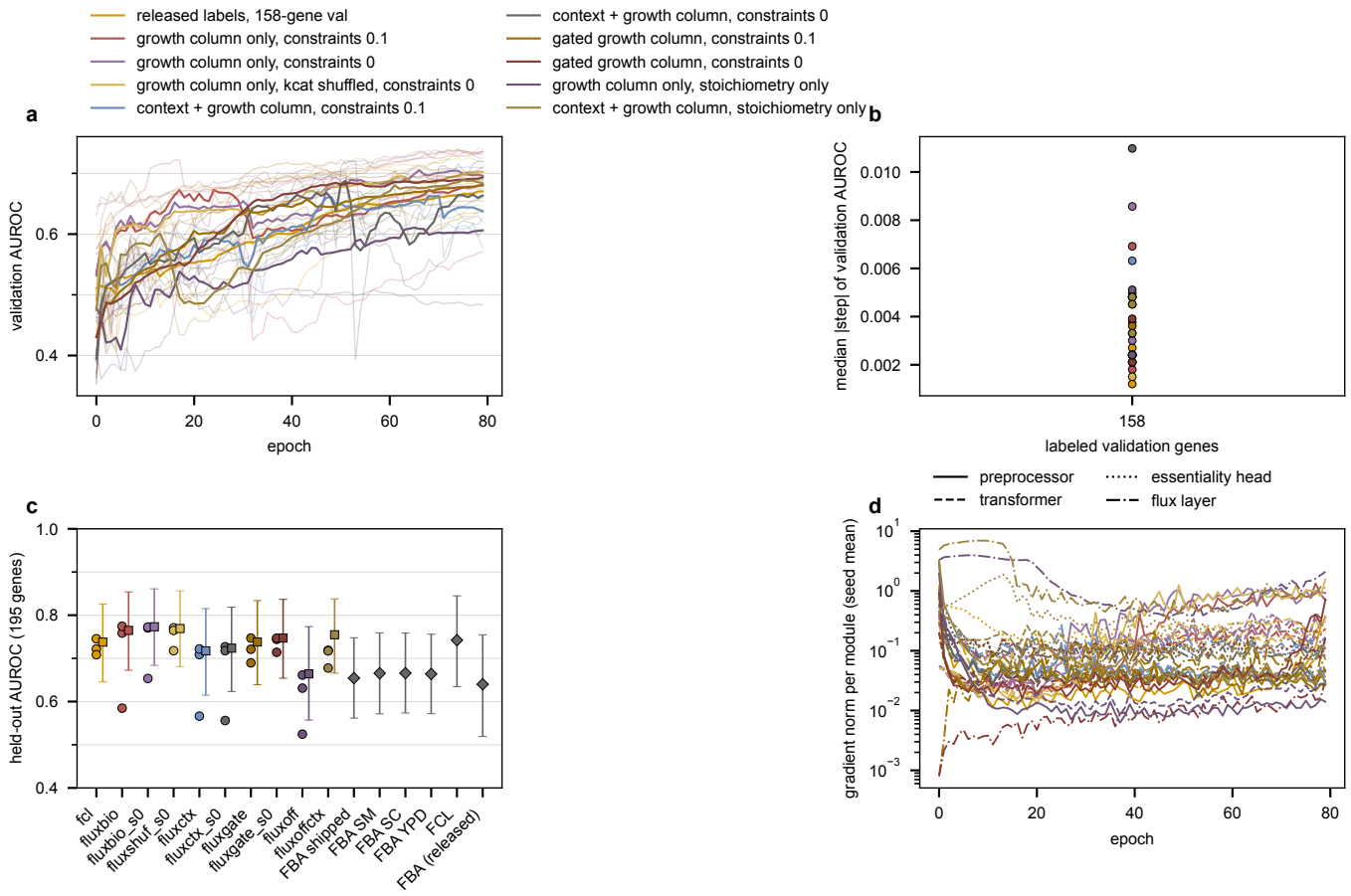


Figure 7: Round 2, the metabolic module. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step of that curve per cell, against the number of labeled validation genes, 158 for every cell in this round. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w13`.

scale 0.1 the same comparison is one of three above. The gate leaves zero and stops: at epoch 80 it reads −0.056 to −0.063 on all six gated cells, and the flux layer's gradient norm sits at 0.01 to 0.07 against 0.22 to 0.36 on the forced arm's two seeds that train. The optimizer takes a small negative weight on the growth affine and the layer receives a tenth of the forced arm's gradient, so the gated cell is a pooled cell with a small correction, and it scores like one.

Help does not hold. The forced arm at scale 0 scores 0.770, 0.653 and 0.772, two of three seeds above the pooled cell and one seed 0.07 below; at scale 0.1 the split is the same and the low seed reads 0.585. The shuffled-k_{cat} control, the same layer with the catalytic constants permuted across units, scores 0.771, 0.718 and 0.765, a seed mean of 0.751 against the real table's 0.732, so whatever the growth column carries here does not come from the correspondence between units and constants. The forced arm's pooled bootstrap, 0.773 [0.68, 0.86], is above Flux Cone Learning's 0.742 and FBA's 0.640 to 0.666 on the same genes, level with the pooled cell's 0.738, and below the ladder's 0.80 line.

The collapse on this endpoint is not the penalty's. Seed 2701 falls at both scales, to 0.585 at 0.1 and 0.653 at 0, and the two runs are the same run: the gradient norms start where the other seeds' do, decay within ten epochs to a hundredth of them (preprocessor 0.002, transformer 0.003 at epoch 80, against 1.0 and 0.5 on seed 2700) and stay there, with the training loss flat at 0.46 against 0.34, which is a stalled seed rather than a starved layer. The stoichiometry-only layer collapses on all three seeds (0.524, 0.631, 0.661), so the capacity bound and the thermodynamic term are what keep the growth column informative rather than what starve it, the opposite of the betaxanthin reading. Adding the context head beside the growth affine without a gate costs 0.06 on the seed mean (0.666 and 0.667 against 0.725) and puts one seed at 0.56; the same sum behind the zero-initialized gate loses nothing.

4.2 Round 4: the module on the best backbone ✗

Round 4 (jobs 2246 to 2248, three cells per GPU) repeats the gated readout, the forced readout and the shuffled-k_{cat} control at constraint scale 0 on the round-3 cell, batch 32 with the class weight at 200 epochs, paired by seed against it (Table 7, Figure 8). The reads are the round-2 reads.

Table 7: Round 4, the module on the round-3 levers. The round-3 pooled cell (batch 32, class weight, 200 epochs) and the same cell with the prediction routed through the flux layer at constraint scale 0: the gated readout, the forced growth column, and the forced growth column with the kcat vector shuffled.

arm	mean	sd	seeds	pooled [95% CI]	selected epochs	val step
batch 32 + class weight, 200 epochs	0.774	0.006	0.780, 0.769, 0.773	0.780 [0.69, 0.87]	75, 68, 148	0.0031
gated growth column, constraints 0 (round-3 levers)	0.771	0.010	0.766, 0.782, 0.765	0.768 [0.68, 0.85]	67, 71, 77	0.0040
growth column only, constraints 0 (round-3 levers)	0.777	0.005	0.772, 0.781, 0.778	0.778 [0.69, 0.86]	108, 74, 84	0.0057
growth column only, kcat shuffled (round-3 levers)	0.779	0.006	0.785, 0.781, 0.772	0.781 [0.69, 0.87]	66, 71, 74	0.0048

Every arm lands within 0.015 of the pooled cell on every seed. The gated readout scores 0.766, 0.782 and 0.765 against 0.780, 0.769 and 0.773, one of three above; the forced readout 0.772, 0.781 and 0.778, two of three above by 0.005 and 0.012 and one below by 0.008; the shuffled-k_{cat} control 0.785, 0.781 and 0.772, two of three above, a seed mean of 0.779 against the real table's 0.777. The gate now moves: it leaves zero after epoch 10 and climbs steadily to −0.17, +0.20 and +0.11 at epoch 200, of either sign, with the flux layer's gradient at 0.06 to 0.12 against 0.22 to 0.65 in the forced arm. The stalled seed of round 2 does not recur under batch 32 with the class weight. On every arm the validation curve peaks between epochs 60 and 110 and falls afterwards (Figure 8a), so the 200-epoch budget is more than the cell uses and the smoothed selection rule is what the number rests on.

The two rounds together answer the two questions. The module does not hurt: on the stable backbone and on the best one, the gated cell is the pooled cell to within seed noise. The module does not help: the forced growth column, which carries the whole prediction, matches the pooled MLP head to within 0.01, and it matches it equally well with the catalytic constants permuted, so the growth column is as expressive a readout as the head but its k_{cat} content is not what it reads. On this endpoint, at this width, the enzyme-constrained flux is a learned nonlinearity over the context rather than a physical prior that ranks genes.

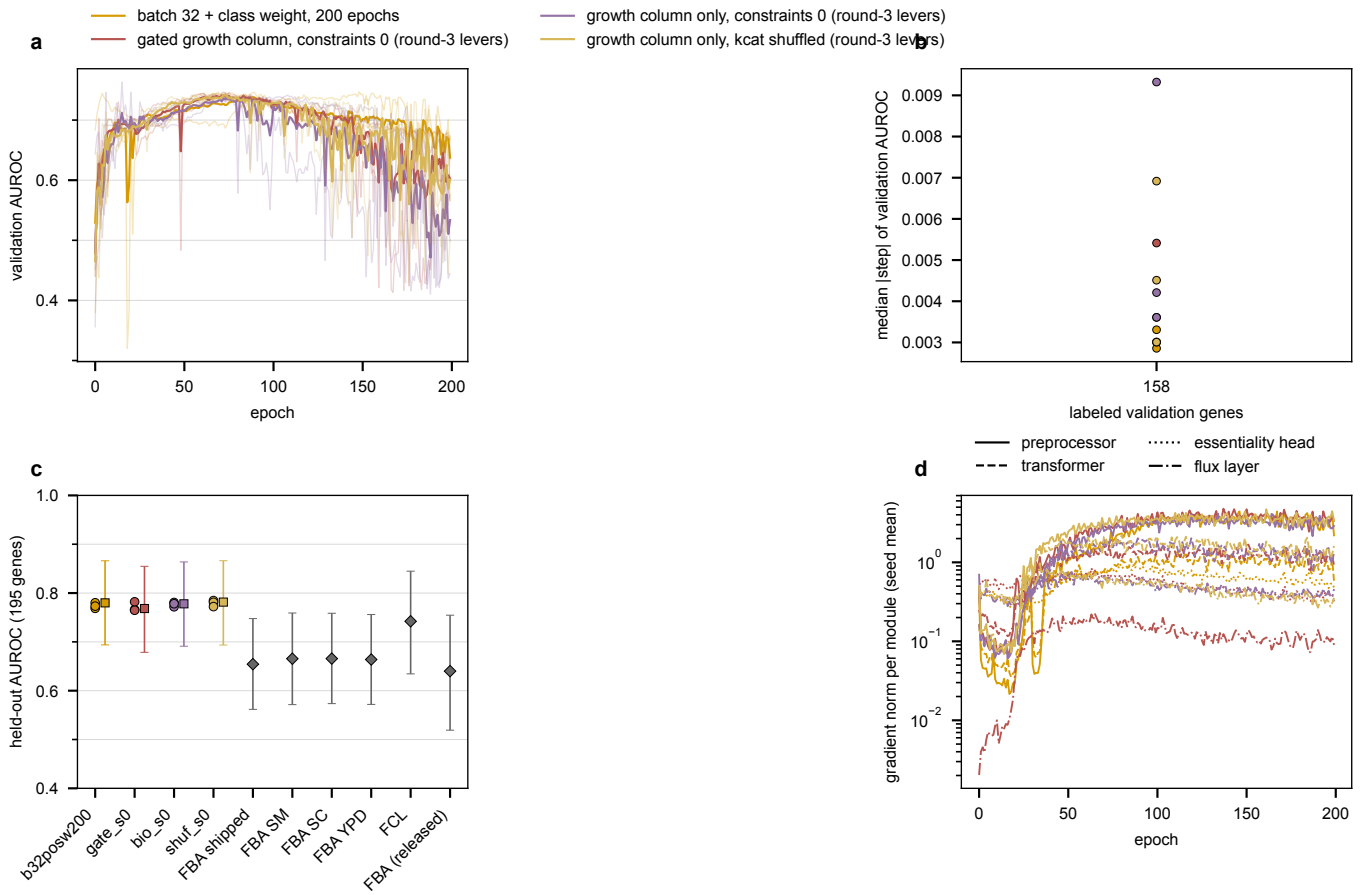


Figure 8: Round 4, the module on the round-3 cell. **a**, Validation AUROC per epoch, one thin line per seed and the seed mean per arm. **b**, The median absolute epoch-to-epoch step per cell against the validation size, 158 for every cell. **c**, Held-out AUROC per cell (points), the seed mean and the gene bootstrap of the pooled per-gene logits (square with interval), beside FBA on four media, Flux Cone Learning and the released FBA predictions on the same 195 genes. **d**, Gradient norm per module per epoch, seed mean; the dash-dot line is the flux layer. Written by `experiments/028-gene-essentiality/scripts/wave12_readout.py -round w15`.

5 Open ✗

- FBA's growth ratio as an input to the transformer. The ladder's top rungs are frozen encodings with the four ratios appended (0.81 to 0.83), and neither half reaches that alone; a cell that concatenates the ratios to the readout, or carries them as a node feature, tests whether the transformer gains the same complement.
- The module's own hyperparameters have not been moved on this endpoint: the k_{cat} tier (measured against predicted), the thermodynamic source, the parameterization (projection against nullspace), the number of alternations.
- Every cell here is width 12. The depth-against-width isolation (width 90 at one block, width 12 at two) has not run on the split-file protocol.